Evaluation Artifacts

Measured retrieval results, transparent evaluation status, test cases, and sample outputs

Retrieval benchmark — 40 queries

Relevance is labeled by same-course membership, a weak proxy rather than advisor relevance judgments. The local release includes synthetic/default data; interpret these scores as exploratory.

Retrieval mode	Precision@5	MRR
Hybrid keyword + TF-IDF/LSA + structured	0.450	0.681
Semantic-only	0.395	0.617
Keyword-only	0.475	0.729

Interpretation

Keyword-only ranks best under this proxy. The result does not establish that hybrid retrieval is superior. Same-course membership can mark profiles relevant even when individual records are not meaningfully similar. See backend/retrieval-evaluation.json for the saved result.

DeepEval scores

Status: NOT RUN; no DeepEval scores are reported. An optional script evaluates Faithfulness, Answer Relevancy, Contextual Relevancy, and custom advisor-support quality using NVIDIA NIM. It is restricted to synthetic profiles. One-record outputs are demonstrations, not representative metrics.

RAGAS scores

Status: NOT RUN; no RAGAS integration or scores are available. Do not treat missing scores as zero or invent values. A run needs a question/context/answer/reference dataset and a configured judge provider.

Test cases

Twenty manual acceptance cases are provided in docs/sample-test-cases.md. They cover authentication and validation, filters, prioritization, similarity, analysis, intervention CRUD, feedback, role controls, ingestion, exports, and provider fallback. These are documented cases, not an automated test pass report.

Sample outputs

deliverables/sample-outputs.json is a deterministic analysis response generated from seeded synthetic profiles only, without an NVIDIA/NIM call. It includes performance evidence, heuristic risk factors, retrieved examples, advisor suggestions, and limitations. Regenerate from backend/ with python scripts/generate_sample_output.py.

Reproducibility and limitations

From backend/, run python scripts/evaluate_retrieval.py to refresh the offline ranking file. To generate optional DeepEval scores, install requirements-evaluation.txt, configure NVIDIA_API_KEY and NVIDIA_MODEL in ignored backend/.env, then run python scripts/evaluate_quality.py. Use only synthetic data for external judge calls. Retrieval labels and one-record evaluations are insufficient for operational or safety claims; obtain advisor-labeled judgments and institutional review.